Description

'''

This dataset contains 55,000 synthetic customer transactions generated using Python's Faker library.

It simulates real-world retail transactions and can be used for data preprocessing, statistical analysis,

and visualization.

Shape of Dataset

Rows: 55,000

Columns: 13

'''

| Column | Description |
|---|---|
| **CID** | Unique Customer ID |
| **TID** | Unique Transaction ID |
| **Gender** | Gender of the customer (Male, Female, Other) |
| **Age Group** | Age category of the customer |
| **Purchase Date** | Date and time when the transaction took place |
| **Product Category** | Category of the purchased product |
| **Discount Availed** | Indicates whether a discount was availed (Yes/No) |
| **Discount Name** | Name of the discount applied |
| **Discount Amount (INR)** | Amount of discount received in Indian Rupees |
| **Gross Amount** | Total amount before applying the discount |
| **Net Amount** | Final amount after applying the discount |
| **Purchase Method** | Payment method used (Credit Card, Debit Card, UPI, etc.) |
| **Location** | City where the purchase occurred |

Code:

```
import pandas as pd

df = pd.read_csv("project1_df.csv")

print(df.shape)
print(df.columns)
print(df.dtypes)
print(df.info())
print(df.head())
```

```
df["Purchase Date"] = pd.to_datetime(
    df["Purchase Date"],
    format="%d/%m/%Y %H:%M:%S"
)
df["Year"] = df["Purchase Date"].dt.year
df["Month"] = df["Purchase Date"].dt.month
df["Day"] = df["Purchase Date"].dt.day
print(df["Purchase Date"].dtypes)
```

```
print(df.isnull().sum())
print((df.isnull().sum()/len(df))*100)
df["Discount Name"] = df["Discount Name"].fillna("No Discount")
print(df.duplicated().sum())
```

```
#statistics
print(df.describe())
print("Mean Gross Amount =", df["Gross Amount"].mean())
print("Mean Net Amount =", df["Net Amount"].mean())
print("Mean Discount Amount =", df["Discount Amount (INR)"].mean())
print("Median of Gross Amount",df["Gross Amount"].median())
print("Median of Net Amount",df["Net Amount"].median())
print("Std of Gross Amount",df["Gross Amount"].std())
print("Std of Net Amount",df["Net Amount"].std())
print("Minimum of Gross Amount",df["Gross Amount"].min())
print("Maximum of Gross amount",df["Gross Amount"].max())
```

```
print(df[["Gross Amount",
          "Discount Amount (INR)",
          "Net Amount"]].corr())
```

[17] 0.1s

| | Gross Amount | Discount Amount (INR) | Net Amount |
|---|---|---|---|
| Gross Amount | 1.000000 | 0.001473 | 0.995400 |
| Discount Amount (INR) | 0.001473 | 1.000000 | -0.094341 |
| Net Amount | 0.995400 | -0.094341 | 1.000000 |

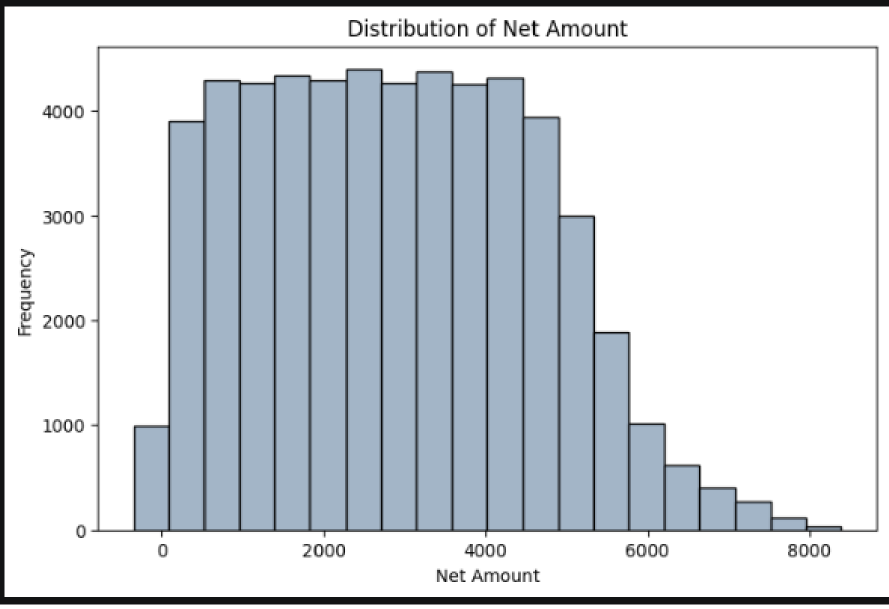
```python
print(df["Gender"].value_counts())
print("\n",df["Purchase Method"].value_counts())
print("\n",df["Location"].value_counts())
```

[19] ✓ 0.0s

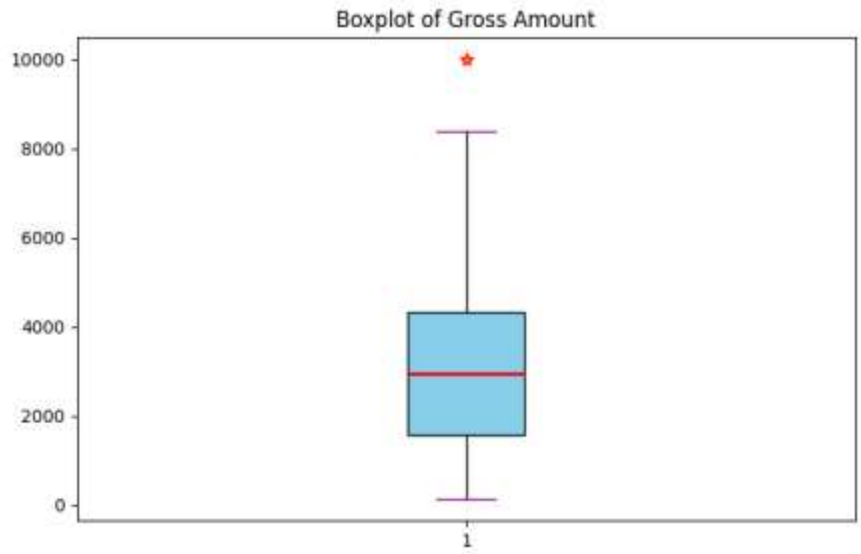
```
Gender
Female    18454
Other     18450
Male      18096
Name: count, dtype: int64

 Purchase Method
Credit Card            22096
Debit Card             13809
Net Banking             5485
International Card      2815
Cash on Delivery        2768
PhonePe UPI             2683
Paytm UPI               2674
Google Pay UPI          2670
Name: count, dtype: int64

 Location
Mumbai       11197
Delhi        10799
Bangalore     8249
Hyderabad     5545
Chennai       4368
Pune          3781
Ahmedabad     2785
Kolkata       2709
Jaipur        1678
Lucknow       1136
Other         1046
Varanasi       606
Dehradun       553
Srinagar       548
Name: count, dtype: int64
```

```python
#plotting
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(df["Net Amount"], bins=20, color="#a3b5c7", edgecolor="black")
plt.xlabel("Net Amount")
plt.ylabel("Frequency")
plt.title("Distribution of Net Amount")
plt.show()
```

[24] ✓ 0.1s




```python
df.loc[0, "Gross Amount"] = 10000
plt.figure(figsize=(8,5))
plt.boxplot(df["Gross Amount"],
            patch_artist=True,
            boxprops=dict(facecolor='skyblue', color='black'),
            medianprops=dict(color='red', linewidth=2),
            whiskerprops=dict(color='black'),
            capprops=dict(color='purple'),
            flierprops=dict(marker='*',
                    markerfacecolor='orange',
                    markersize=8,
                    markeredgecolor='red'))
plt.title("Boxplot of Gross Amount")
plt.show()
Q1 = df["Gross Amount"].quantile(0.25)
Q2 = df["Gross Amount"].quantile(0.50)
Q3 = df["Gross Amount"].quantile(0.75)
IQR = Q3 - Q1

print("Lower limit =", Q1 - 1.5*IQR)
print("Upper limit =", Q3 + 1.5*IQR)
print("Q1 =", Q1)
print("Median =", Q2)
print("Q3 =", Q3)
```




Lower limit = -2607.981562499999

Upper limit = 8512.5991375

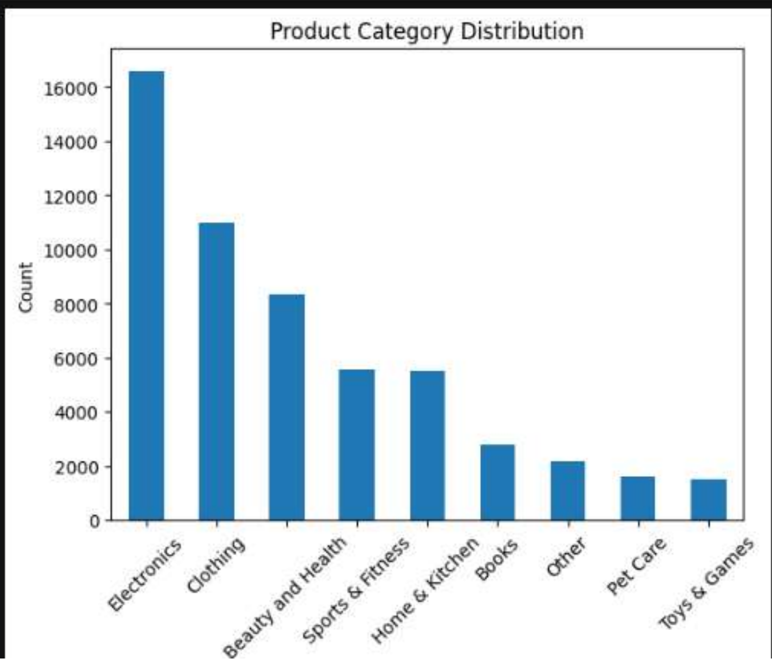
Q1 = 1562.2362

Median = 2954.4383500000004

Q3 = 4342.381375

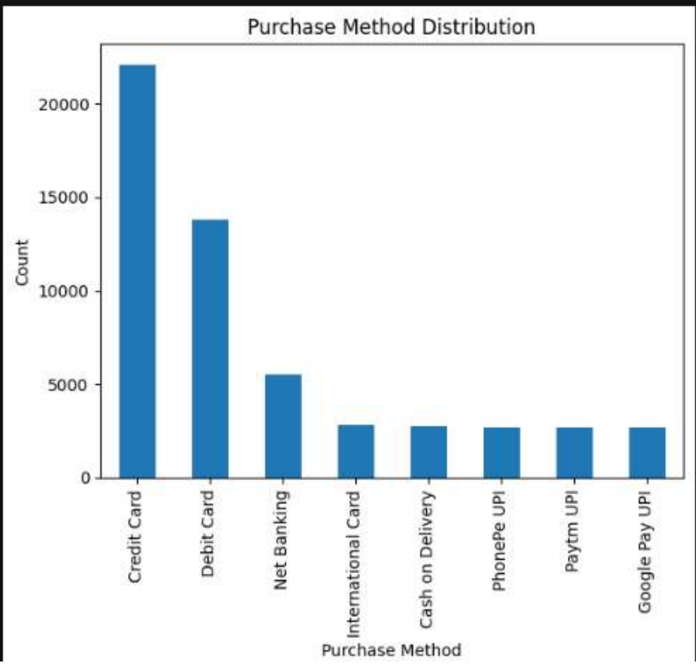
```
df["Product Category"].value_counts().plot(kind='bar')

plt.xlabel("Product Category")
plt.ylabel("Count")
plt.title("Product Category Distribution")
plt.xticks(rotation=45)
plt.show()
```

[26] ✓ 0.1s



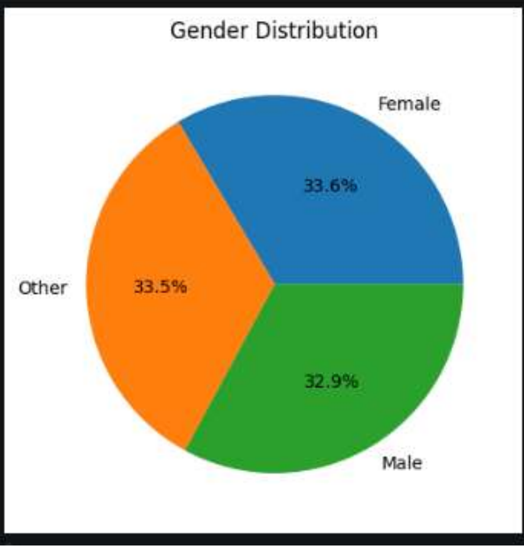
```python
df["Purchase Method"].value_counts().plot(kind='bar')

plt.xlabel("Purchase Method")
plt.ylabel("Count")
plt.title("Purchase Method Distribution")
plt.show()
```

[27] ✓ 0.1s




```python
df["Gender"].value_counts().plot(kind='pie',
                                 autopct='%1.1f%%')

plt.ylabel("")
plt.title("Gender Distribution")
plt.show()
```

[28] ✓ 0.1s



```python
df["Location"].value_counts().head(10).plot(kind='bar')

plt.xlabel("City")
plt.ylabel("Number of Purchases")
plt.title("Top 10 Locations")
plt.xticks(rotation=45)
plt.show()
```

[29] ✓ 0.2s